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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HACKATHON 2026

Idea Submission Template — Code. Create. Innovate.

Team Name:

Team Raksha

Problem Statement No:

2

Problem Statement Title:

Improving Safety at Large Public Events

Team Leader Name:

Shounak Joshi

IDEA TITLE

Know the crowd. Know the person. Stop the surge.

Cameras watch. The software remembers faces.

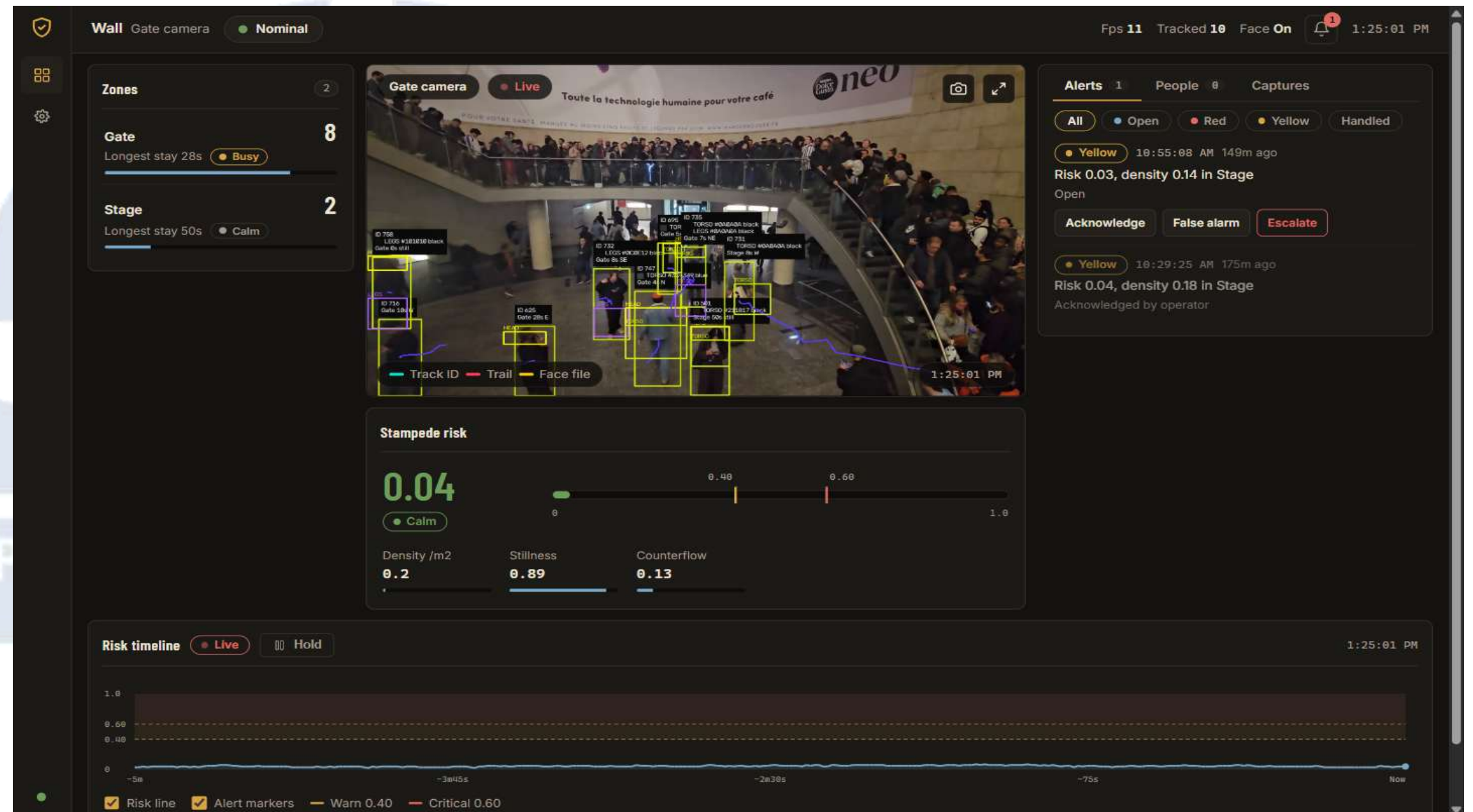
It warns guards before a crush builds.

0.4s <1s 15

face recognised guard warned live pictures a second

Watch it work:

1. A face appears on camera.
2. Their file pops up in 0.4s.
3. A crowded zone turns yellow in under 1s.

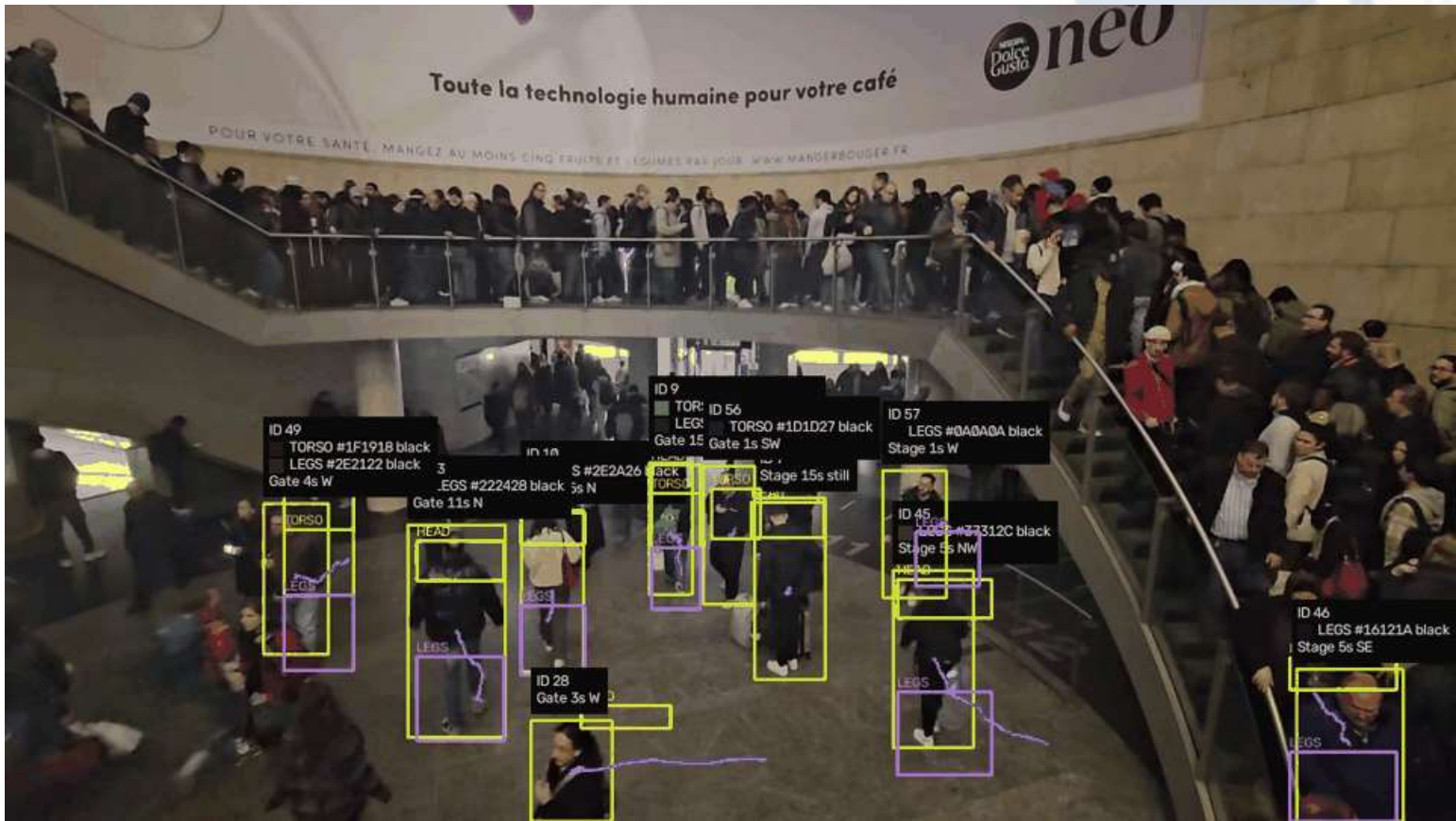


Above: the live wall on real station footage.

TECHNICAL APPROACH



In plain words: the camera watches, the software follows every person, and it shouts before a crush builds.



Only good photos kept

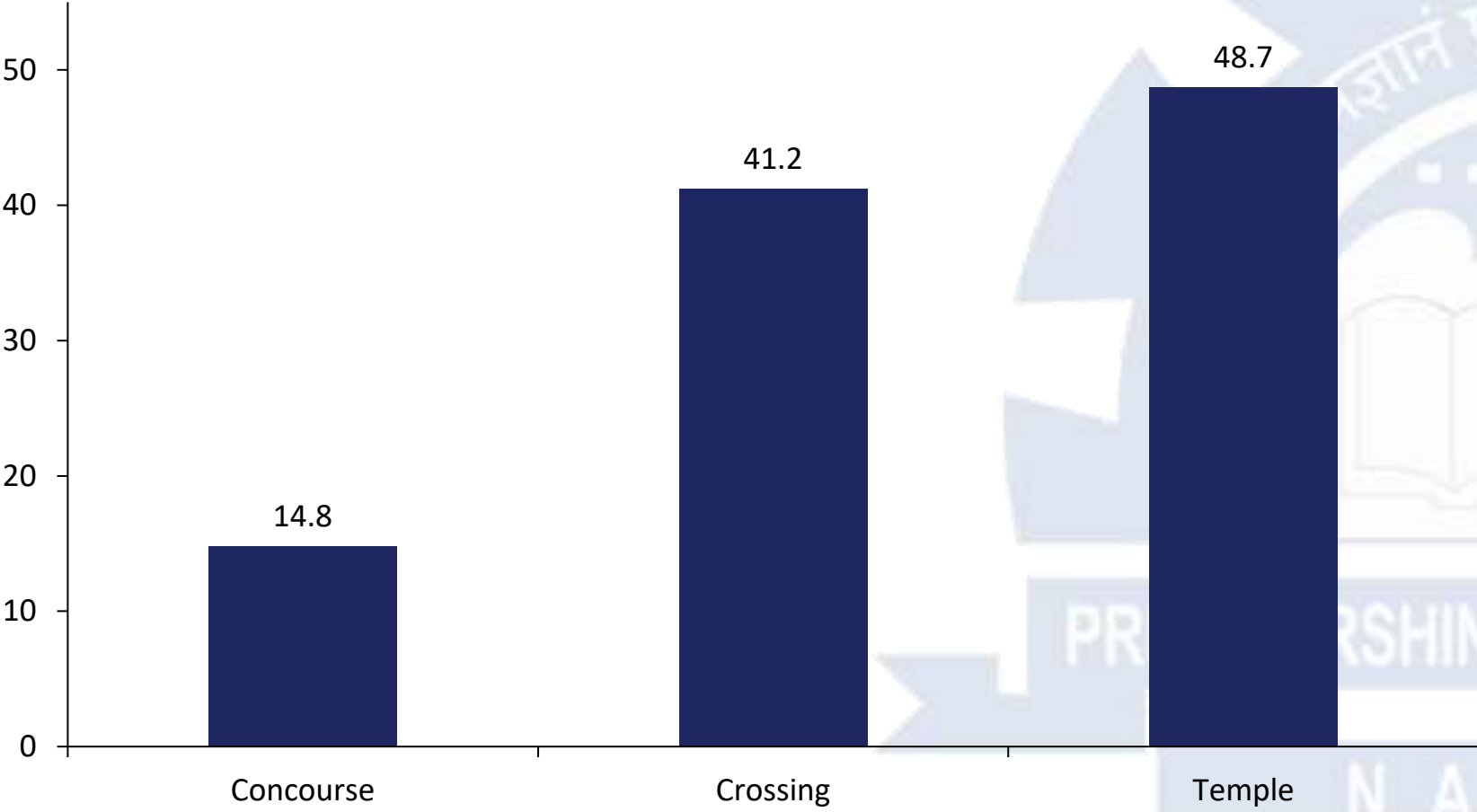
Only sharp front photos join the memory. Blurry ones are ignored, so matches stay honest.

Tested on real video

18 people followed at once. 850 people held across 2 minutes of station video.

FEASIBILITY AND VIABILITY

How fast it watches (higher is better)



One gaming laptop. The heaviest crowd still runs smooth.

Instant

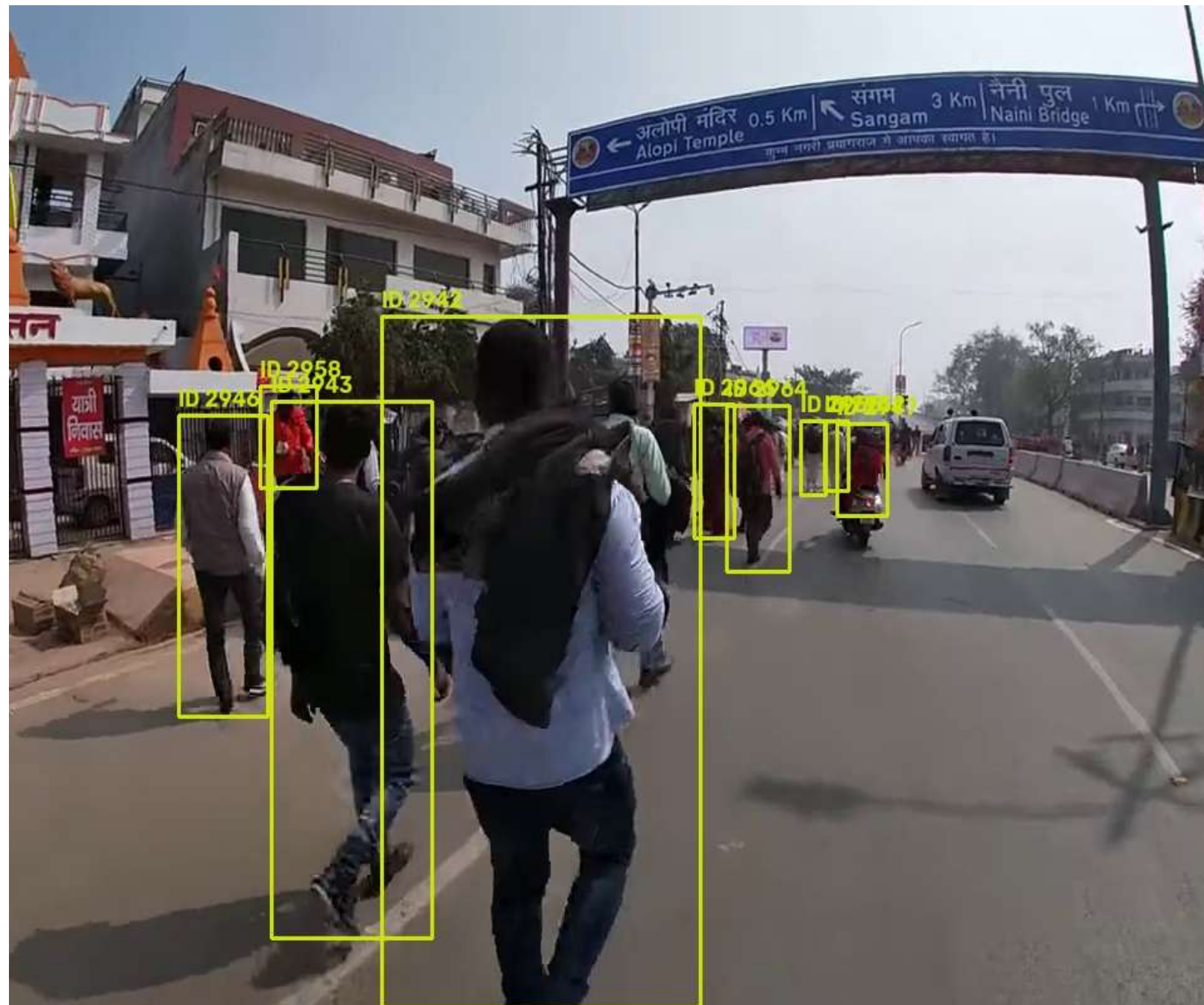
a face matched in 0.06 seconds

Red alert

fires when a crowd packs tight and stops moving



IMPACT AND BENEFITS



Before: 50 screens, one tired guard, nothing gets flagged.

After: one wall watches all cameras and warns only the risky zones.

Organizers

Crowd jam spotted 60 seconds before people notice. Guards told which zone to clear.

Police

Find a lost child or suspect by photo in under 2 seconds. Full history in one click.

Public

Safer exits, faster help. Faces stay blurred in storage.

RESEARCH AND REFERENCES

People tracking: YOLOv8 plus ByteTrack. 2025-26 studies call it the best speed balance.

Face matching: ArcFace plus Faiss. Scores 99.5 percent on the standard face test.

Crush science: AURORA and FTLE models. A Delhi trial cut response time 37 percent.

Why India: Hathras, Kumbh, station crushes all came from packed halls plus blocked exits.

Blurred storage

faces blurred on disk

7-day purge

auto delete, no backlog

Audit log

hash-chained actions

No ID linkage

opt-in gallery only